

PROPRIETARY AND CONFIDENTIAL



Matching Freddie Mac Credit Data to Identify HARP Loans

August 1, 2013

Loans that have been refinanced through Home Affordable Refinance Program (HARP) generally prepay slower than other refinance loans that have undergone full underwriting. Correct identification of HARP loans is therefore important. For loans that have a loan-to-value (LTV) ratio above 80%, the level of private mortgage insurance (PMI) coverage may be used to identify an HARP loan. Until recently there was no way to effectively identify HARP loans with LTV ratios of 80% or lower. In March 2013, Freddie Mac provided loan-level credit information for fully underwritten 30-year fixed-rate mortgages originated from January 1999 through December 2011, that it had securitized or purchased. The credit dataset provided an indirect way to identify 30-year fixed-rate HARP loans with any LTV ratio.

In this study, we focus on the loans in the credit dataset that were originated between June 2009 and December 2011. Using the provided loan-level attributes, we match the loans in the credit dataset to the loans in the standard Freddie Mac loan level prepayment database. Non-cash-out refinance loans that are not matched in the standard database are assumed to be HARP loans.

We validate the matching results by examining the prepayment behavior of the HARP loans with LTV ratio less or equal to 80%.

Out of the approximately 15.7 million loans whose credit performance data was released by Freddie Mac, 2,332,419 loans were originated between June 2009 and December 2011. We matched 2,311,921, or just over 99% of these loans, leaving 20,498 unmatched. The goal is to match 100% of the loans so we will continue to explore ways of improving the match.

In order to identify HARP loans, we restrict the matching to the standard 30-year fixed-rate loans with FGLMC ticker. The total FGLMC loan count for loans originated between June 2009 and December 2011 was 3,064,439. Out of these, 2,294,202 loans were matched to the credit data, leaving 770,137 as potential HARP loans. The HARP loan count reduced to 735,046 by retaining only non-cash-out refinance loans.

Focusing on HARP loans with LTV ratio of 80% or lower, we find that these loans generally have lower loan sizes, lower FICO, and higher or unknown DTI. This may explain why these HARP loans prepay slower than standard refinance loans. However, after accounting for loan size, DTI level, and SATO, HARP loans prepay slower than comparable non-HARP loans.

15,699,483 mortgage loans meeting the following criteria:

- Mortgages originated from January 1, 1999, through December 31, 2011, that were sold to Freddie Mac or issued in Freddie Mac participation certificates
- Fully amortizing 30-year fixed-rate mortgages
- Mortgages categorized as having verified or waived documentation (collectively, mortgages with “full documentation”)

The following types of mortgage loans were excluded:

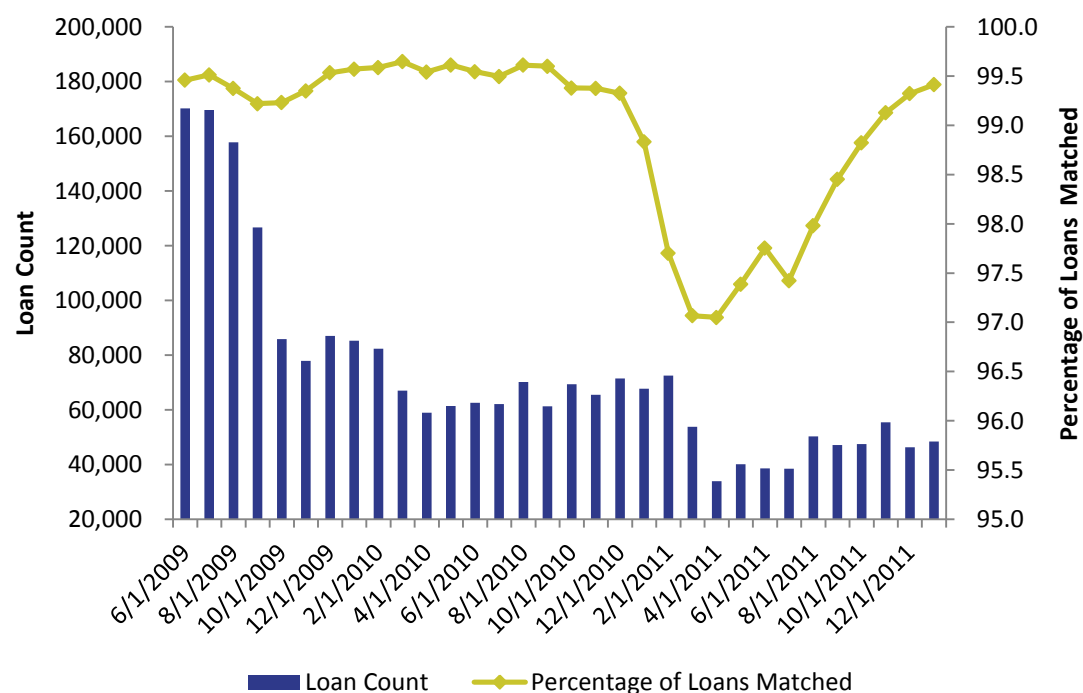
- Adjustable rate mortgages, initial interest, balloon, and any mortgage with step rates
- Relief refinance mortgages (including HARP)
- Government insured mortgages (FHA, VA, etc.)
- Special affordable mortgage products such Home Possible Mortgages
- Mortgages delivered to Freddie Mac under alternate agreements
- Mortgages for which documentation was not verified or not waived
- Mortgages associated with Mortgage Revenue Bonds purchased by Freddie Mac
- Mortgages delivered to Freddie Mac with credit enhancement other than primary mortgage insurance, with exception of certain lender-negotiated credit enhancements
- Mortgages with original loan terms of 420 months or more, and 300 months or less

The new Freddie Mac loan level credit data is first matched to the standard Freddie Mac loan level database. Since the goal is to eventually identify HARP loans, we focus on loans originated between June 2009 and December 2011 in the credit data set – a total of 2,332,419 loans.

The figure to the right shows the number of loans per first payment date from the credit data set, and the percentage of the loans that are matched to the standard database. The lowest percentage matched for any first payment date is 97%, but in aggregate only 20,498 loans, or just under 1%, are not matched.

Any loan in the standard database that is not matched to a loan from the credit file is a potential HARP loan. This means that any credit loan that is not matched results in a potential wrong classification as a HARP loan. It is therefore important to get a matching rate of 100%.

Freddie Mac Credit Data Loan Count by First Payment Date, and Percentage of the Loans that are Matched



To identify HARP loans originated between June 2009 and December 2011, we focus on 30-year fixed-rate loans with a ticker of “FGLMC” in the standard Freddie Mac database. There were 3,064,349 such loans. Out of these, 2,294,202 loans were matched to the new credit dataset, and were not HARP loans. Of the remaining 770,237 loans, 35,191 were cash-out refinance or purchase loans, leaving only 735,046 as HARP loans.

386,217 of these HARP loans had original LTV ratio of 80% or lower. These loans could not have been identified without the new Freddie Mac loan-level credit dataset.

348,829 HARP loans had original LTV ratio above 80%. PMI based mapping had 337,074 of these loans correctly mapped as MHA, giving an error rate of 3.37% for the old mapping method.

Matching of FGLMC Loans Originated from June 2009 to December 2011

Total Number of FGLMC Loans	Loans Matched to Credit Dataset	Unmatched Loans - Potential HARP Loans	HARP Loans (Non-Cashout Refinance)
3,064,439	2,294,202	770,237	735,046

Loan Count of HARP Loans by Original LTV

Original LTV <= 80%	Original LTV > 80%	ALL HARP Loans
386,217	348,829	735,046

Comparison of New vs Old MHA Mapping for Original LTV > 80%

New Mapping Loan Count	Old Mapping Loan Count	Old Mapping Error Rate
348,829	337,074	3.37%

The table compares the characteristics of Freddie Mac 30-year fixed-rate HARP and non-HARP loans with original LTV of 80% or lower, and origination years of 2010 and 2011.

For both 2010 and 2011 origination years, HARP loans have higher gross wac, higher retail percentage, lower FICO, and lower average and weighted average original loan sizes.

These characteristics imply that the HARP loans are likely to prepay slower than non-HARP loans, and it is important to determine if the slower prepayment is due to other attributes independent of the HARP status.

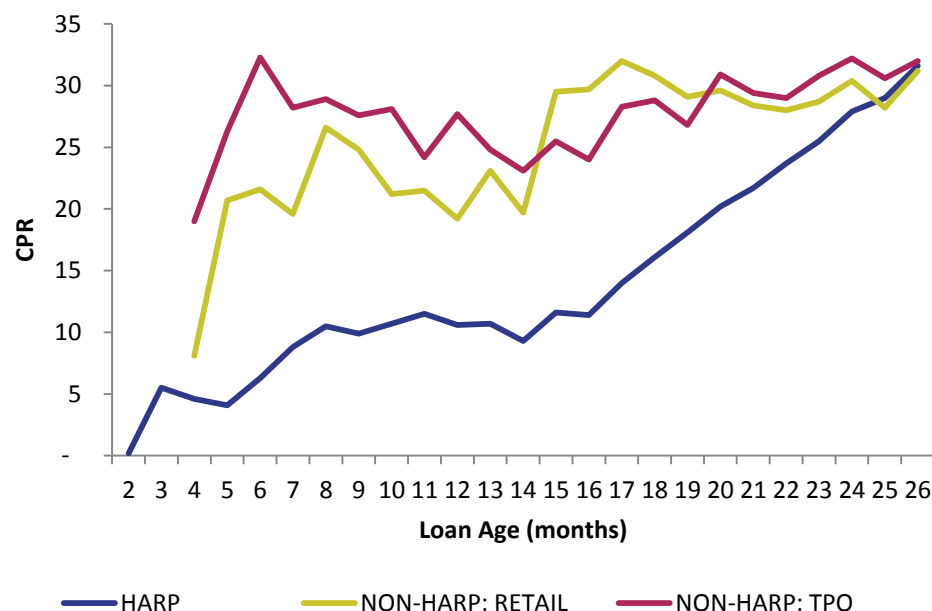
Charateristics of FH 30-Year Non Cash-out Refi Loans with OLTV <= 80%

Vintage	HARP Status	Issue Amt (\$BB)	OLTV	Gwac	Pct Retail	FICO	WAOLS	AOLS
2010	HARP	35.08	63	4.95	95	756	255,147	202,190
2010	NON-HARP	61.12	65	4.83	41	766	324,578	249,223
2011	HARP	25.71	62	4.69	95	755	241,067	186,854
2011	NON-HARP	42.02	65	4.48	35	766	325,957	255,283

The figure shows seasoning curves for HARP and non-HARP loans with original LTV ratio of 80% or lower, and originated in 2010 and 2011. Only loans with original balance between 200,000 and 350,000, and rate incentive between 0.5% and 2.0% are considered. Prepayment data is from the first half of 2012.

Below 15 months of seasoning, non-HARP TPO prepay approximately 3-8 cpr faster than non-HARP retail. HARP loans prepay about 10 cpr slower than non-HARP retail. The prepayments converge as seasoning increases; at 26 months, the prepayments are comparable.

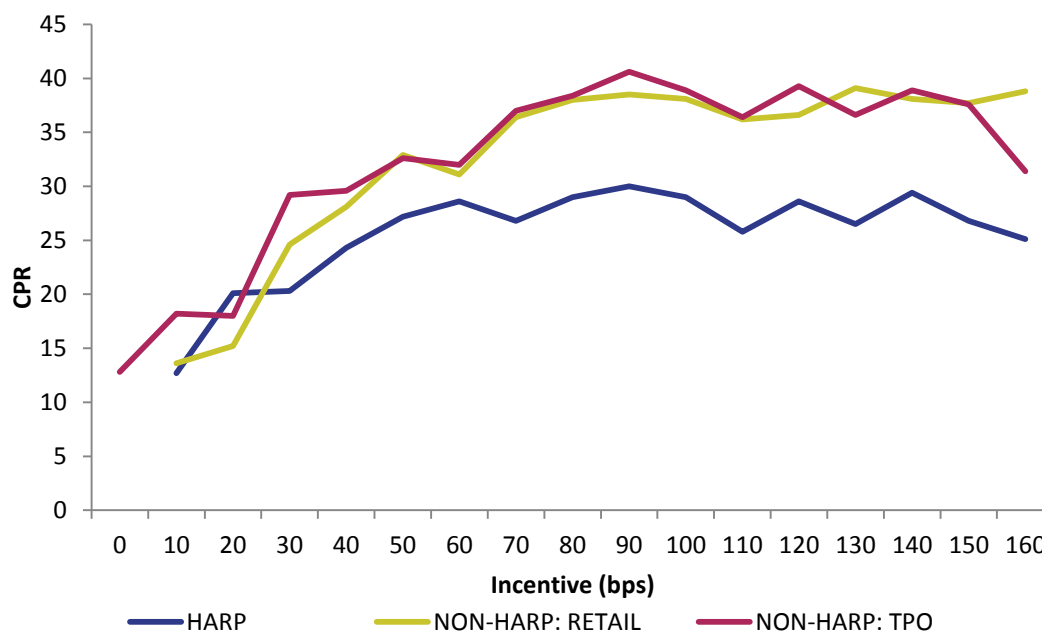
Seasoning Curve by HARP and Origination Channel Using 1H 2012 Prepayment



The figure shows S-Curves for HARP and non-HARP loans originated in 2010 and 2011 with original LTV ratio less or equal to 80%, and original loan balance between 200,000 and 350,000. Loan age is restricted to 12-36 months and the prepayment data is from 2013.

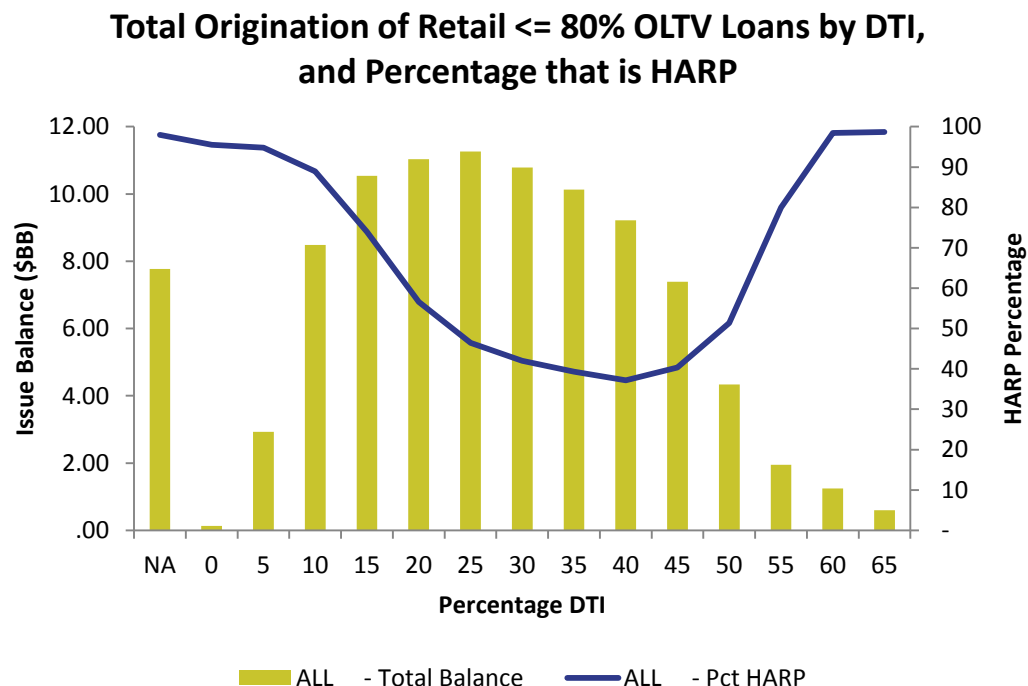
HARP loans are about 10 CPR slower than non-HARP loans. For the loan age considered, non-HARP TPO and non-HARP retail prepay at comparable rates.

S-Curve for ≤ 80 OLTV 12-36 Months Seasoned Non-Cashout Refi Loans by HARP and Origination Channel Using 2013 Prepayments



The figure shows the total issue balance by DTI percentage of retail loans originated in 2010 and 2011 that had original LTV ratio of 80% or lower. Also shown is the percentage of the issue balance that is HARP.

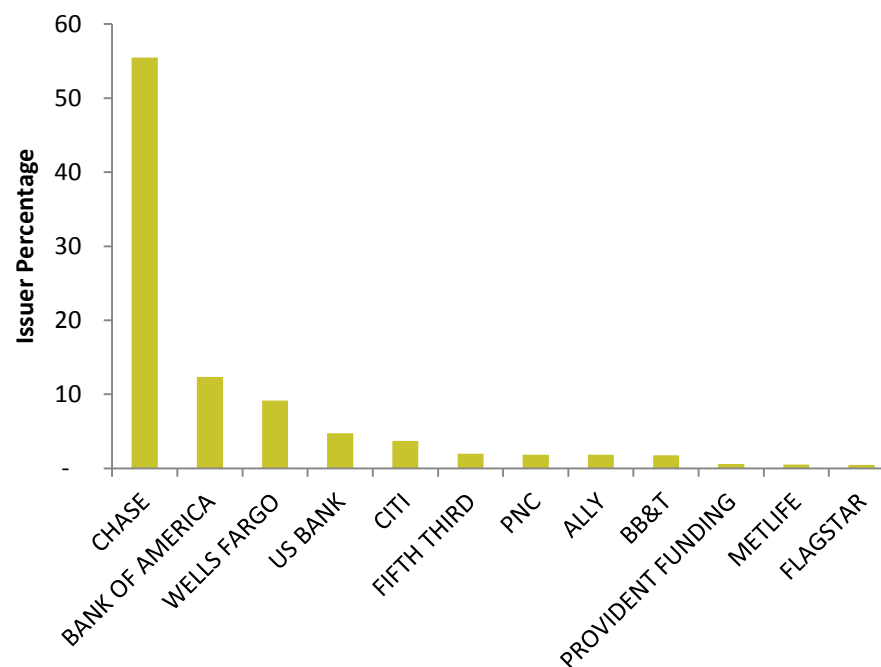
Majority of loans with DTI of 45 and higher, or with unknown ("NA") DTI, are HARP loans.



The figure shows the percentage of HARP retail loans with either unknown DTI or DTI of 45 or higher that were issued by a particular issuer. The loans were originated in 2010 and 2011 and they had original LTV of 80% or lower.

Chase issued the majority of loans with high or unknown DTI.

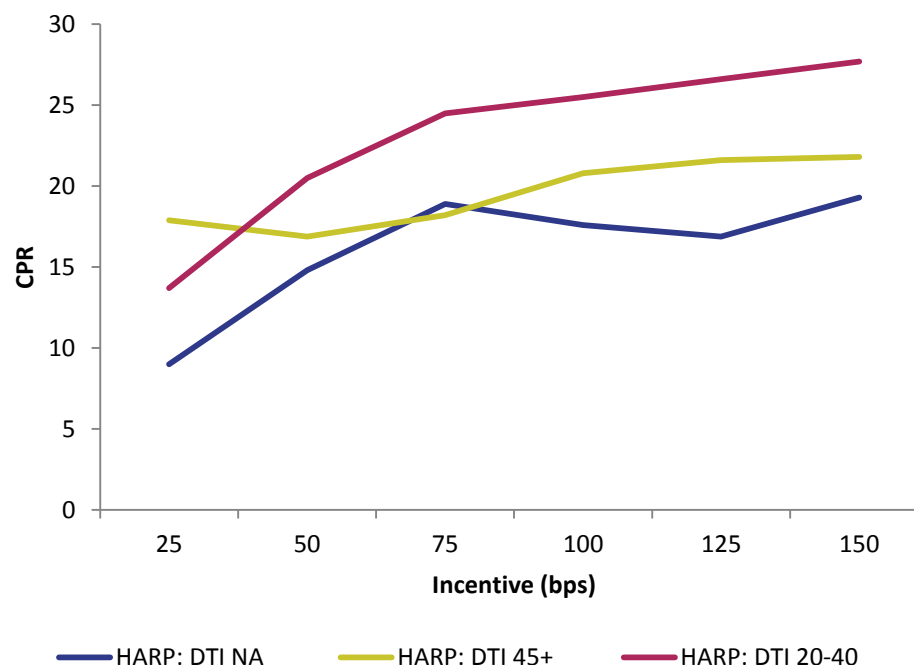
Issuer Percentage for Loans that have DTI of 45 and Above or Unknown DTI



The figure shows S-curves for HARP loans by DTI buckets. The curves are based on retail loans originated in 2010 and 2011 that had original balance between 200,000 and 350,000, and original LTV of 80% or less. The prepayment data is from the second half of 2012 through July 2013.

HARP loans with unknown DTI pay the slowest, followed by loans with DTI of 45 and above, and loans with DTI between 20-40 are the fastest.

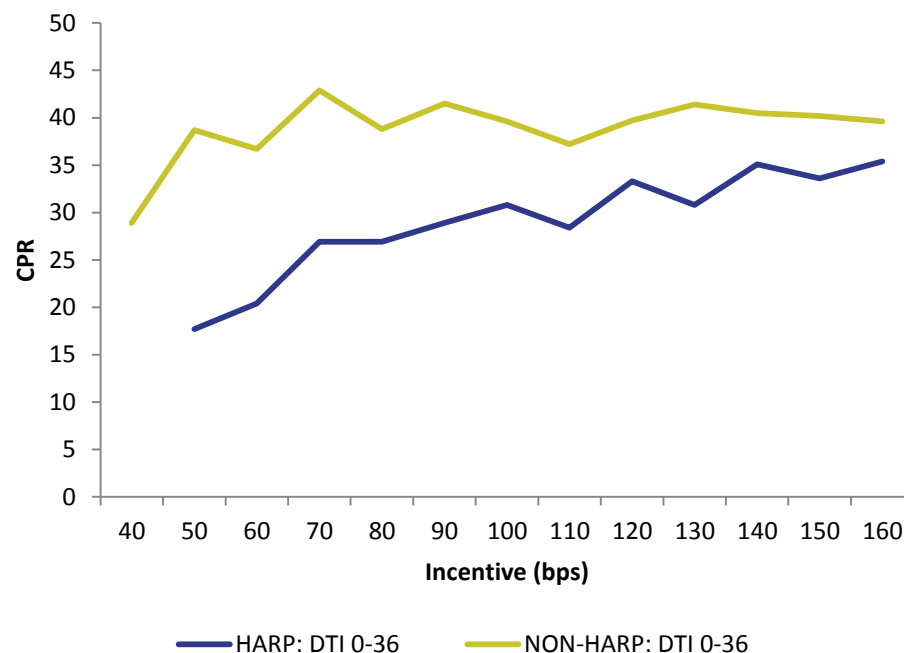
S-Curves by DTI Buckets for HARP Loans Using 2H of 2012 and 2013 Prepayments



The figure shows S-curves for HARP and non-HARP loans with DTI between 0 and 36. The curves are based on retail loans originated in 2010 and 2011 that had original balance between 200,000 and 350,000, and original LTV ratio of 80% or lower. The prepayment data is from the second half of 2012 through July 2013.

HARP loans are slower than non-HARP loans but the prepayment speeds converge as incentive increases.

HARP and Non-HARP S-Curves for DTI Between 0 and 36%, and SATO Between -12.5 and 12.5 bps



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